LAZARD LEGACY MATH ACADEMY

Statistics & Probability

Quick Drill — 30 Minutes

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"Dreams can come true. Better late than never."

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Section 1: Quick Lesson — Mean, Median, Mode, Range

Definitions:

• **Mean (Average):** Add all values, then divide by the number of values. Mean = Sum ÷ Count.

• **Median:** The middle value when all numbers are arranged in order. If there's an even number of values, average the two middle numbers.

• **Mode:** The value that appears most frequently. A data set can have no mode, one mode, or multiple modes.

• **Range:** The difference between the highest and lowest values. Range = Max - Min.

Steps for Finding Each Measure:

1. First, arrange data from least to greatest.

2. For Mean: add all numbers and divide by count.

3. For Median: find the middle position (if n values, middle is at position (n+1)/2).

4. For Mode: look for the most repeated number.

5. For Range: subtract smallest from largest.

Worked Examples:

Example 1:

Data set: {4, 8, 6, 5, 9, 8, 3}. Find the mean.

Solution: Sum = 4 + 8 + 6 + 5 + 9 + 8 + 3 = 43. Count = 7. Mean = 43 ÷ 7 = **6.14** (rounded to hundredths).

Example 2:

Data set: {12, 5, 8, 3, 15, 9, 7}. Find the median.

Solution: Step 1: Order the data: 3, 5, 7, 8, 9, 12, 15. Step 2: Count = 7, so middle is the 4th value. Median = **8**.

Example 3:

Data set: {2, 5, 3, 5, 8, 5, 9, 1}. Find the mode.

Solution: Count each value: 1→once, 2→once, 3→once, 5→three times, 8→once, 9→once. The number 5 appears most often. Mode = **5**.

Example 4:

Data set: {14, 22, 8, 31, 17}. Find the range.

Solution: Maximum = 31. Minimum = 8. Range = 31 - 8 = **23**.

Example 5:

Test scores: {85, 90, 78, 92, 85, 88}. Find mean, median, and mode.

Solution: Mean = (85+90+78+92+85+88) ÷ 6 = 518 ÷ 6 = **86.3**. Ordered: 78, 85, 85, 88, 90, 92. Median = (85+88) ÷ 2 = **86.5**. Mode = **85** (appears twice).

Example 6:

Data: {10, 15, 10, 20, 25, 10, 30}. Find all four measures.

Solution: Ordered: 10, 10, 10, 15, 20, 25, 30. Mean = 120 ÷ 7 = **17.14**. Median = **15** (4th value). Mode = **10** (appears 3 times). Range = 30 - 10 = **20**.

Section 2: Quick Lesson — Basic Probability

Key Formula:

P(event) = Number of favorable outcomes ÷ Total number of outcomes

Important Facts:

- Probability always ranges from 0 (impossible) to 1 (certain).
- Can be written as a fraction, decimal, or percentage.
- P(not A) = 1 - P(A). This is the complement rule.
- For independent events: P(A and B) = P(A) × P(B).

Common Probability Scenarios:

- Coin flip: P(heads) = 1/2
- Standard die: P(any single number) = 1/6
- Deck of cards: 52 cards, 4 suits of 13, P(any suit) = 13/52 = 1/4

Worked Examples:

Example 1:

A bag has 3 red, 5 blue, and 2 green marbles. What is P(blue)?

Solution: Total marbles = 3 + 5 + 2 = 10. Favorable (blue) = 5. P(blue) = 5/10 = **1/2 = 0.5 = 50%**.

Example 2:

A fair die is rolled. What is P(rolling a 4)?

Solution: Total outcomes = 6 (numbers 1–6). Favorable = 1 (just the 4). P(4) = 1/6 ≈ **0.167 or about 16.7%**.

Example 3:

A coin is flipped twice. What is P(two heads)?

Solution: All outcomes: HH, HT, TH, TT (4 total). Favorable = 1 (only HH). P(two heads) = 1/4 = **0.25 = 25%**.

Example 4:

A standard deck of 52 cards. What is P(drawing a heart)?

Solution: Hearts in deck = 13. Total cards = 52. P(heart) = 13/52 = 1/4 = **0.25 = 25%**.

Example 5:

A spinner has 8 equal sections: 3 red, 2 blue, 2 green, 1 yellow. Find P(not red).

Solution: Total sections = 8. Red = 3. Not red = 8 - 3 = 5. P(not red) = 5/8 = **0.625 = 62.5%**.

Section 3: Quick Lesson — Reading Graphs & Tables

Tips for Reading Data Displays:

• Always read the title, axis labels, and legend first.

• For bar graphs: the height of each bar represents the value.

• For pie charts: each slice is a percentage of the whole (total = 100%).

• For tables: read across rows and down columns carefully.

• Watch for units! (thousands, millions, percentages)

Worked Examples:

Example 1:

A bar graph shows daily sales: Mon=12, Tue=8, Wed=15, Thu=10, Fri=20. What is the total for the week?

Solution: Add all values: 12 + 8 + 15 + 10 + 20 = **65 total sales**.

Example 2:

Using the same data (Mon=12, Tue=8, Wed=15, Thu=10, Fri=20), what is the average daily sales?

Solution: Total = 65. Number of days = 5. Average = 65 ÷ 5 = **13 sales per day**.

Example 3:

A table shows student grades: A=5, B=12, C=8, D=3, F=2. What percentage of students earned a B?

Solution: Total students = 5 + 12 + 8 + 3 + 2 = 30. Students with B = 12. Percentage = 12/30 = 0.40 = **40%**.

Example 4:

A pie chart shows preferences: 25% Sports, 35% Music, 20% Art, 20% Reading. If 200 students were surveyed, how many chose Music?

Solution: Music = 35% of 200. Calculate: 0.35×200 = **70 students** chose Music.

Section 4: Practice Drill — 35 Problems

Directions: Solve each problem. Show your work in the space provided.

Mean, Median, Mode, Range (1–10):

1. Find the mean: {7, 12, 9, 15, 3, 8}.

2. Find the median: {22, 14, 31, 18, 25}.

3. Find the mode: {4, 7, 4, 9, 2, 4, 8, 7}.

4. Find the range: {56, 23, 89, 45, 12, 67}.

5. Data: {10, 20, 30, 40, 50}. Find mean and median.

6. Test scores: {72, 85, 91, 68, 85, 77}. Find mean, median, and mode.

7. Data: {3, 3, 5, 7, 7, 7, 9, 11}. Find all four measures.

8. Data: {100, 85, 90, 95, 80}. Find the mean.

9. Data: {6, 2, 8, 4, 10, 12, 6}. Find the median.

10. Data: {1, 1, 2, 3, 3, 3, 4, 5}. Find the mode.

Probability (11–25):

11. A bag has 4 red, 6 blue, 5 green balls. What is P(green)?

12. A die is rolled. P(even number)?

13. A die is rolled. P(number greater than 4)?

14. A coin flipped 3 times. P(all tails)?

15. A standard 52-card deck. P(king)?

16. A standard 52-card deck. P(red card)?

17. Spinner: 5 equal sections numbered 1-5. P(odd)?

18. A bag has 8 red, 2 white marbles. P(not white)?

19. If 40 students took a test and 30 passed, what is P(pass)?

20. A survey: 15 chose pizza, 10 burgers, 5 tacos. P(pizza)?

21. Two dice rolled. P(sum = 7)?

22. A class of 25: 10 boys, 15 girls. P(selecting a girl)?

23. A spinner has 10 sections: 4 red, 3 blue, 2 green, 1 yellow. P(red or blue)?

24. Two coins flipped. P(at least one head)?

25. A bag: 6 red, 4 blue. You draw 1. P(red)?

Graphs, Tables & Mixed (26–35):

26. A bar graph shows: Math=85, Science=90, English=78, History=82. What is the average score?

27. Data: {55, 60, 65, 70, 75, 80, 85}. Find median and range.

28. Data: {4, 4, 4, 8, 8, 12}. Find mean and mode.

29. Test grades: {95, 87, 92, 88, 95, 80}. Find the mode.

30. Data: {5, 10, 15, 20, 25}. Find mean and median.

31. P(drawing a face card from a standard deck)?

32. Data: {18, 22, 18, 25, 30, 18, 27}. Find all four measures.

33. Heights (in): {60, 62, 65, 63, 68, 61, 64}. Find the mean.

34. Data: {7, 7, 8, 9, 10, 11, 12, 12}. Find the median.

35. A pie chart shows 40% prefer Brand A out of 500 people surveyed. How many prefer Brand A?

Section 5: Answer Key

1. Mean = 54÷6 = 9

2. Median = 22 (ordered: 14,18,22,25,31)

3. Mode = 4 (appears 3 times)

4. Range = 89-12 = 77

5. Mean = 150÷5 = 30, Median = 30

6. Mean = 478÷6 ≈ 79.7, Median = (77+85)÷2 = 81, Mode = 85

7. Mean = 52÷8 = 6.5, Median = 7, Mode = 7, Range = 11-3 = 8

8. Mean = 450÷5 = 90

9. Median = 6 (ordered: 2,4,6,6,8,10,12)

10. Mode = 3 (appears 3 times)

11. P = 5/15 = 1/3 ≈ 33.3%

12. P = 3/6 = 1/2 = 50%

13. P = 2/6 = 1/3 ≈ 33.3%

14. P = $(1/2)^3$ = 1/8 = 12.5%

15. P = 4/52 = 1/13 ≈ 7.7%

16. P = 26/52 = 1/2 = 50%

17. P = 3/5 = 60%

18. P = 8/10 = 4/5 = 80%

19. P = 30/40 = 3/4 = 75%

20. P = 15/30 = 1/2 = 50%

21. P = 6/36 = 1/6 ≈ 16.7%

22. P = 15/25 = 3/5 = 60%

23. P = 7/10 = 70%

24. P = 3/4 = 75%

25. P = 6/10 = 3/5 = 60%

26. Average = 335÷4 = 83.75

27. Median = 65, Range = 85-55 = 30

28. Mean = 40÷6 ≈ 6.67, Mode = 4

29. Mode = 95 (appears twice)

30. Mean = 75÷5 = 15, Median = 15

31. P = 12/52 = 3/13 ≈ 23.1%

32. Mean ≈ 22.6, Median = 22, Mode = 18, Range = 30-18 = 12

33. Mean = 443÷7 ≈ 63.3

34. Median = (9+10)÷2 = 9.5

35. 40% × 500 = 200 people

STUDENT NOTES